



PHY101 important Physics Formulae

An Introduction to Physics (Virtual University of Pakistan)



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TABLE OF INFORMATION FOR 2002

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ADVANCED PLACEMENT PHYSICS B EQUATIONS FOR 2002

NEWTONIAN MECHANICS

$$v = v_0 + at$$

$$x = x_0 + v_0 t + \frac{1}{2} at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$\Sigma \mathbf{F} = \mathbf{F}_{net} = m\mathbf{a}$$

$$F_{fric} \leq \mu N$$

$$a_c = \frac{v^2}{r}$$

$$\tau = rF \sin \theta$$

$$\mathbf{p} = m\mathbf{v}$$

$$\mathbf{J} = \mathbf{F}\Delta t = \Delta \mathbf{p}$$

$$K = \frac{1}{2} mv^2$$

$$\Delta U_g = mgh$$

$$W = \mathbf{F} \cdot \Delta \mathbf{r} = F\Delta r \cos \theta$$

$$P_{avg} = \frac{W}{\Delta t}$$

$$P = \mathbf{F} \cdot \mathbf{v} = Fv \cos \theta$$

$$\mathbf{F}_s = -k\mathbf{x}$$

$$U_s = \frac{1}{2} kx^2$$

$$T_s = 2\pi\sqrt{\frac{m}{k}}$$

$$T_p = 2\pi\sqrt{\frac{\ell}{g}}$$

$$T = \frac{1}{f}$$

$$F_G = -\frac{Gm_1m_2}{r^2}$$

$$U_G = -\frac{Gm_1m_2}{r}$$

a = acceleration
 F = force
 f = frequency
 h = height
 J = impulse
 K = kinetic energy
 k = spring constant
 ℓ = length
 m = mass
 N = normal force
 P = power
 p = momentum
 r = radius or distance
 $\mathbf{r}$ = position vector
 T = period
 t = time
 U = potential energy
 v = velocity or speed
 W = work done on a system
 x = position
 μ = coefficient of friction
 θ = angle
 τ = torque

ELECTRICITY AND MAGNETISM

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1q_2}{r^2}$$

$$\mathbf{E} = \frac{\mathbf{F}}{q}$$

$$U_E = qV = \frac{1}{4\pi\epsilon_0} \frac{q_1q_2}{r}$$

$$E_{avg} = -\frac{V}{d}$$

$$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$$

$$C = \frac{Q}{V}$$

$$C = \frac{\epsilon_0 A}{d}$$

$$U_c = \frac{1}{2} QV = \frac{1}{2} CV^2$$

$$I_{avg} = \frac{\Delta Q}{\Delta t}$$

$$R = \frac{\rho \ell}{A}$$

$$V = IR$$

$$P = IV$$

$$C_p = \sum_i C_i$$

$$\frac{1}{C_s} = \sum_i \frac{1}{C_i}$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

$$F_B = qvB \sin \theta$$

$$F_B = BIl \sin \theta$$

$$B = \frac{\mu_0}{2\pi} \frac{I}{r}$$

$$\phi_m = \mathbf{B} \cdot \mathbf{A} = BA \cos \theta$$

$$\mathcal{E}_{avg} = -\frac{\Delta \phi_m}{\Delta t}$$

$$\mathcal{E} = B\ell v$$

A = area
 B = magnetic field
 C = capacitance
 d = distance
 E = electric field
 $\mathcal{E}$ = emf
 F = force
 I = current
 ℓ = length
 P = power
 Q = charge
 q = point charge
 R = resistance
 r = distance
 t = time
 U = potential (stored) energy
 V = electric potential or potential difference
 v = velocity or speed
 ρ = resistivity
 ϕ_m = magnetic flux

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ADVANCED PLACEMENT PHYSICS B EQUATIONS FOR 2002

FLUID MECHANICS AND THERMAL PHYSICS

$$p = p_0 + \rho gh$$

$$F_{buoy} = \rho Vg$$

$$A_1 v_1 = A_2 v_2$$

$$p + \rho gy + \frac{1}{2} \rho v^2 = \text{const.}$$

$$\Delta \ell = \alpha \ell_0 \Delta T$$

$$Q = mL$$

$$Q = mc\Delta T$$

$$p = \frac{F}{A}$$

$$pV = nRT$$

$$K_{avg} = \frac{3}{2} k_B T$$

$$v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3k_B T}{\mu}}$$

$$W = -p\Delta V$$

$$Q = nc\Delta T$$

$$\Delta U = Q + W$$

$$\Delta U = nc_V \Delta T$$

$$e = \left| \frac{W}{Q_H} \right|$$

$$e_c = \frac{T_H - T_C}{T_H}$$

A = area

c = specific heat or molar specific heat

e = efficiency

F = force

h = depth

K_{avg} = average molecular kinetic energy

L = heat of transformation

ℓ = length

M = molecular mass

m = mass of sample

n = number of moles

p = pressure

Q = heat transferred to a system

T = temperature

U = internal energy

V = volume

v = velocity or speed

v_{rms} = root-mean-square velocity

W = work done on a system

y = height

α = coefficient of linear expansion

μ = mass of molecule

ρ = density

WAVES AND OPTICS

$$v = f\lambda$$

$$n = \frac{c}{v}$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\sin \theta_c = \frac{n_2}{n_1}$$

$$\frac{1}{s_i} + \frac{1}{s_o} = \frac{1}{f}$$

$$M = \frac{h_i}{h_o} = -\frac{s_i}{s_o}$$

$$f = \frac{R}{2}$$

$$d \sin \theta = m\lambda$$

$$x_m \approx \frac{m\lambda L}{d}$$

d = separation

f = frequency or focal length

h = height

L = distance

M = magnification

m = an integer

n = index of refraction

R = radius of curvature

s = distance

v = speed

x = position

λ = wavelength

θ = angle

GEOMETRY AND TRIGONOMETRY

Rectangle

$$A = bh$$

Triangle

$$A = \frac{1}{2}bh$$

Circle

$$A = \pi r^2$$

$$C = 2\pi r$$

Parallelepiped

$$V = \ell wh$$

Cylinder

$$V = \pi r^2 \ell$$

$$S = 2\pi r \ell + 2\pi r^2$$

Sphere

$$V = \frac{4}{3}\pi r^3$$

$$S = 4\pi r^2$$

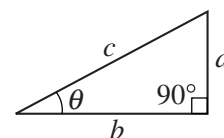
Right Triangle

$$a^2 + b^2 = c^2$$

$$\sin \theta = \frac{a}{c}$$

$$\cos \theta = \frac{b}{c}$$

$$\tan \theta = \frac{a}{b}$$



ATOMIC AND NUCLEAR PHYSICS

$$E = hf = pc$$

$$K_{\max} = hf - \phi$$

$$\lambda = \frac{h}{p}$$

$$\Delta E = (\Delta m)c^2$$

E = energy

f = frequency

K = kinetic energy

m = mass

p = momentum

λ = wavelength

ϕ = work function

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ADVANCED PLACEMENT PHYSICS C EQUATIONS FOR 2002

MECHANICS

$$\begin{aligned}
 v &= v_0 + at & a &= \text{acceleration} \\
 x &= x_0 + v_0 t + \frac{1}{2} at^2 & F &= \text{force} \\
 v^2 &= v_0^2 + 2a(x - x_0) & f &= \text{frequency} \\
 \sum \mathbf{F} &= \mathbf{F}_{\text{net}} = m\mathbf{a} & h &= \text{height} \\
 \mathbf{F} &= \frac{d\mathbf{p}}{dt} & I &= \text{rotational inertia} \\
 \mathbf{J} &= \int \mathbf{F} dt = \Delta \mathbf{p} & J &= \text{impulse} \\
 \mathbf{p} &= m\mathbf{v} & K &= \text{kinetic energy} \\
 F_{\text{fric}} &\leq \mu N & k &= \text{spring constant} \\
 W &= \int \mathbf{F} \cdot d\mathbf{r} & \ell &= \text{length} \\
 K &= \frac{1}{2} mv^2 & L &= \text{angular momentum} \\
 P &= \frac{dW}{dt} & m &= \text{mass} \\
 P &= \mathbf{F} \cdot \mathbf{v} & N &= \text{normal force} \\
 \Delta U_g &= mgh & P &= \text{power} \\
 a_c &= \frac{v^2}{r} = \omega^2 r & p &= \text{momentum} \\
 \boldsymbol{\tau} &= \mathbf{r} \times \mathbf{F} & r &= \text{radius or distance} \\
 \sum \boldsymbol{\tau} &= \boldsymbol{\tau}_{\text{net}} = I\boldsymbol{\alpha} & \mathbf{r} &= \text{position vector} \\
 I &= \int r^2 dm = \sum mr^2 & T &= \text{period} \\
 \mathbf{r}_{\text{cm}} &= \sum m\mathbf{r} / \sum m & t &= \text{time} \\
 v &= r\omega & U &= \text{potential energy} \\
 \mathbf{L} &= \mathbf{r} \times \mathbf{p} = I\boldsymbol{\omega} & v &= \text{velocity or speed} \\
 K &= \frac{1}{2} I\omega^2 & W &= \text{work done on a system} \\
 \omega &= \omega_0 + \alpha t & x &= \text{position} \\
 \theta &= \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2 & \mu &= \text{coefficient of friction} \\
 \mathbf{F}_s &= -k\mathbf{x} & \theta &= \text{angle} \\
 U_s &= \frac{1}{2} kx^2 & \tau &= \text{torque} \\
 T &= \frac{2\pi}{\omega} = \frac{1}{f} & \omega &= \text{angular speed} \\
 T_s &= 2\pi \sqrt{\frac{m}{k}} & \alpha &= \text{angular acceleration} \\
 T_p &= 2\pi \sqrt{\frac{\ell}{g}} \\
 \mathbf{F}_G &= -\frac{Gm_1 m_2}{r^2} \hat{\mathbf{r}} \\
 U_G &= -\frac{Gm_1 m_2}{r}
 \end{aligned}$$

ELECTRICITY AND MAGNETISM

$$\begin{aligned}
 F &= \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} & A &= \text{area} \\
 \mathbf{E} &= \frac{\mathbf{F}}{q} & B &= \text{magnetic field} \\
 \oint \mathbf{E} \cdot d\mathbf{A} &= \frac{Q}{\epsilon_0} & C &= \text{capacitance} \\
 E &= -\frac{dV}{dr} & d &= \text{distance} \\
 V &= \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i} & E &= \text{electric field} \\
 U_E &= qV = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} & \mathcal{E} &= \text{emf} \\
 C &= \frac{Q}{V} & F &= \text{force} \\
 C &= \frac{\kappa \epsilon_0 A}{d} & I &= \text{current} \\
 C_p &= \sum_i C_i & L &= \text{inductance} \\
 \frac{1}{C_s} &= \sum_i \frac{1}{C_i} & \ell &= \text{length} \\
 I &= \frac{dQ}{dt} & n &= \text{number of loops of wire per unit length} \\
 U_c &= \frac{1}{2} QV = \frac{1}{2} CV^2 & P &= \text{power} \\
 R &= \frac{\rho \ell}{A} & Q &= \text{charge} \\
 V &= IR & q &= \text{point charge} \\
 R_s &= \sum_i R_i & R &= \text{resistance} \\
 \frac{1}{R_p} &= \sum_i \frac{1}{R_i} & r &= \text{distance} \\
 P &= IV & t &= \text{time} \\
 \mathbf{F}_M &= q\mathbf{v} \times \mathbf{B} & U &= \text{potential or stored energy} \\
 \oint \mathbf{B} \cdot d\boldsymbol{\ell} &= \mu_0 I & V &= \text{electric potential} \\
 \mathbf{F} &= \int I d\boldsymbol{\ell} \times \mathbf{B} & v &= \text{velocity or speed} \\
 B_s &= \mu_0 nI & \rho &= \text{resistivity} \\
 \phi_m &= \int \mathbf{B} \cdot d\mathbf{A} & \phi_m &= \text{magnetic flux} \\
 \mathcal{E} &= -\frac{d\phi_m}{dt} & \kappa &= \text{dielectric constant} \\
 \mathcal{E} &= -L \frac{dI}{dt} \\
 U_L &= \frac{1}{2} LI^2
 \end{aligned}$$

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Rectangle

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Sphere

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Right Triangle

$$a^2 + b^2 = c^2$$

$$\sin \theta = \frac{a}{c}$$

$$\cos \theta = \frac{b}{c}$$

$$\tan \theta = \frac{a}{b}$$

A = area

C = circumference

V = volume

S = surface area

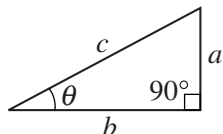
b = base

h = height

ℓ = length

w = width

r = radius



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CALCULUS

$$\frac{df}{dx} = \frac{df}{du} \frac{du}{dx}$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1$$

$$\int e^x dx = e^x$$

$$\int \frac{dx}{x} = \ln|x|$$

$$\int \cos x dx = \sin x$$

$$\int \sin x dx = -\cos x$$